MONASH University



DISTRIBUTED WIRELESS SENSOR NETWORK

OBJECTIVES

- Analyse the design principles of distributed parallel computing systems (ULO3)
- Research and solve synchronisation problems typical in the design of parallel applications (ULO5)

MARKS

- The lab session is worth 14 of the unit's total mark.

LAB INSTRUCTIONS

1. Preparation is required for this Lab/Applied session. Please **DO NOT** plan to complete the session without any preparation/understanding.
2. Students should start working on the tasks at least 1 week prior to the assessed session.
3. **All students** must submit their answers and associated codes via Moodle in the required format before the start of the allocated class. A **late penalty** (5% per day) will apply if you do not submit the required materials on time! Students in the same team/group should submit the same set of files.
4. In addition to regular offline/off-class marking, students' (or Team's) slides, answers, and/or source code will also be reviewed and assessed through in-class presentations, peer reviews, and/or question-and-answer sessions.
5. For **team/group presentations**, marks in general will be allocated for:
    - The correctness of the design, code, or results
    - The quality and clarity of the presentation
    - The correctness of the explanation during the presentation

    The presentation content should include slides that are clear, easy to understand, and readable in a professional working environment (aka an industry setting). While the teaching team *does not impose* strict limits on the presentation format (slides/document format), nevertheless, if the presentation is hard to read/digest (e.g., speech is misaligned with the presented content, or slides/speech are unclear, or arguments are

made without sound logic), marks will be deducted. Both the quality and correctness of your submitted/demonstrated work are assessed and graded.

6. For **individual/group interviews and Q&A sessions**, marks will be allocated for:
    - The correctness of the verbal answer
    - The quality of the verbal answer

    Focus on the most important idea in the answer. Marks will be awarded on the student's ability to explain their ideas with reference to the submitted code or documents in a concise, focused and accurate manner. Answers with errors, inaccurate or lengthy explanations will lose marks. Showing little to no understanding of the code will result in a loss of marks. Showing little to no understanding of the associated libraries or submitted results will result in a loss of marks. Showing little to no understanding of how to select and filter information (aka dumping unfiltered answers without processing from the textbook/internet/AI agent) will lose marks.
7. Always make sure you read and follow the marking rubric well in advance of the deadline.
8. Marks will not be awarded if you skip the class, or do not make any submissions, or do not make any contributions in the working group/team, or make an empty submission to Moodle (unless a special consideration extension has been approved).
9. You are allowed to use Generative-AI to search for information and resources during the preparation period. However, you must declare in your report and upload all the prompt records (in PDF files).
10. AI tools are not allowed during the presentation period or any oral/coding interview sessions.
11. All submitted files should ideally include students' names, ID and Monash email addresses.

LAB IN-CLASS ASSESSMENT ACTIVITIES

Students are required to form teams of 2.

Note: For students who cannot form teams due to exceptional circumstances, arrangements must be made with the teaching team (generally the lead staff member in the corresponding applied/lab class) on/before Week 11.

- Team Preparation Period (before deadline): Divide the workload among team members and complete all the tasks. For each team, you must submit the required files (e.g., report, source code, slides, and corresponding documents) before the deadline.

- Team Presentation Period (max 7 mins): For each team, you are required to present and demonstrate your design, code, results and observations during the session. All team members must be present in the lab/applied session.

- Q&A Period (2 min): Each team member will be asked a few questions based on the submitted and/or presented work.

  Note: If the required submissions are missing or any member(s) of the student team are absent, no marks will be given to the assessment.

Overview

In Week 6 Applied #1, you were required to design an architecture to simulate a wireless sensor network (WSN) of a set of interconnected EV charging nodes.

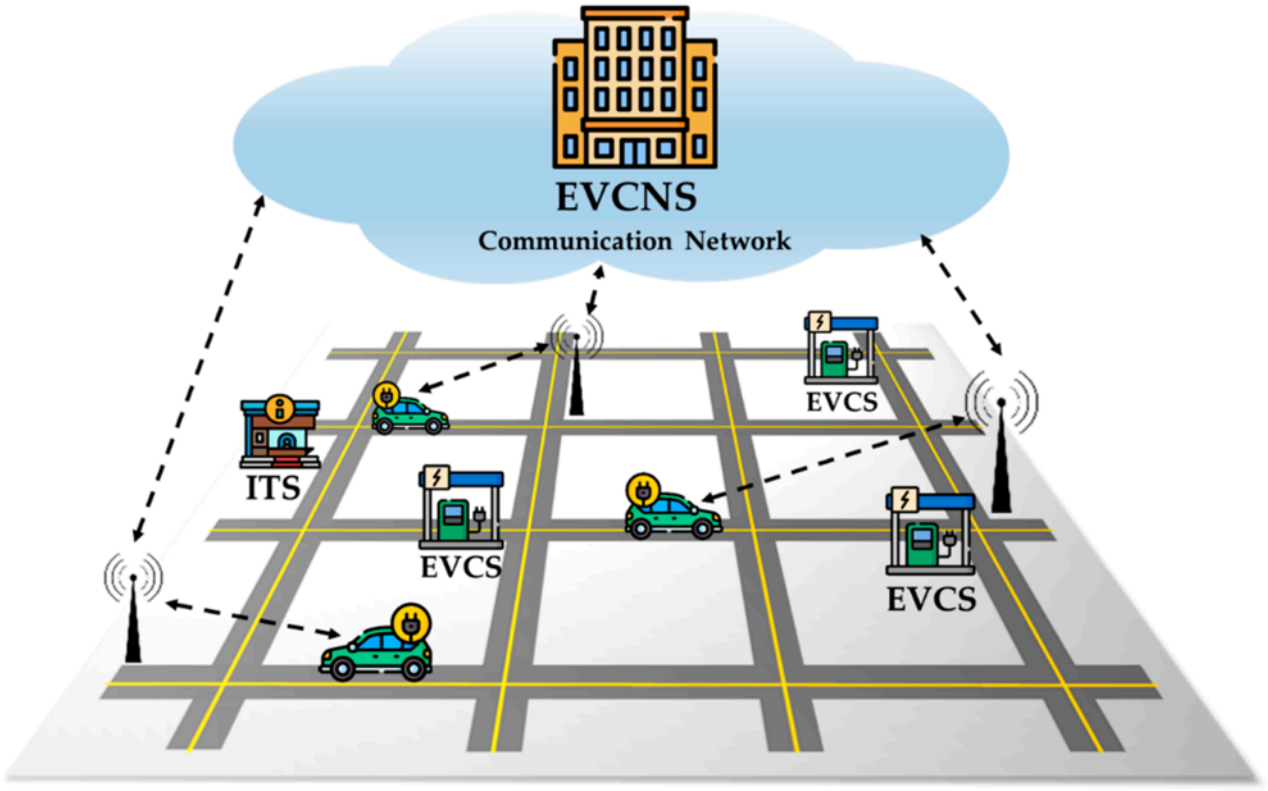


Figure 1. An electric vehicle charging navigation system (EVCNS) which consists of the EV charging station (EVCS) and intelligent transport system (ITS) that monitors the real-time traffic condition. Image sourced from https://www.mdpi.com/1996-1073/13/23/6255

Figure 1 illustrates an EV charging navigation system that consists of an intelligent transport system (ITS) and the EV charging stations. Data collected from EV charging stations could be transmitted to nearby EV charging navigation system centres or to a base station for further processing.

In your previous assessment, the charging nodes are positioned so that each node communicates with its immediate neighbours. Each EV charging node has a number of in-use or free charging ports. If the majority of the ports (over the threshold) are in full use, the node will prompt for neighbour node data. If the received data from all the neighbours also indicates that their ports are heavily utilised (over the threshold), then the original node will alert the base station that the node and all its quadrants are nearly used up.

The base station will receive data from the nodes and log it. The base station will also be able to compute the closest available charging node (shortest distance from the original node) and alert that node to redirect all future incoming vehicles to it.

The aim of the previous assessment (Week 6 Applied #2) was to design an architecture to simulate the wireless sensor network (WSN) setup using the **Message Passing Interface (MPI)**. **In this assessment (Week 12 Lab #3), you are required to implement your architecture (with any improvements/changes/modifications, if necessary) in C/C++ that would simulate the wireless sensor network (WSN) setup with Open MPI and Open MP.**

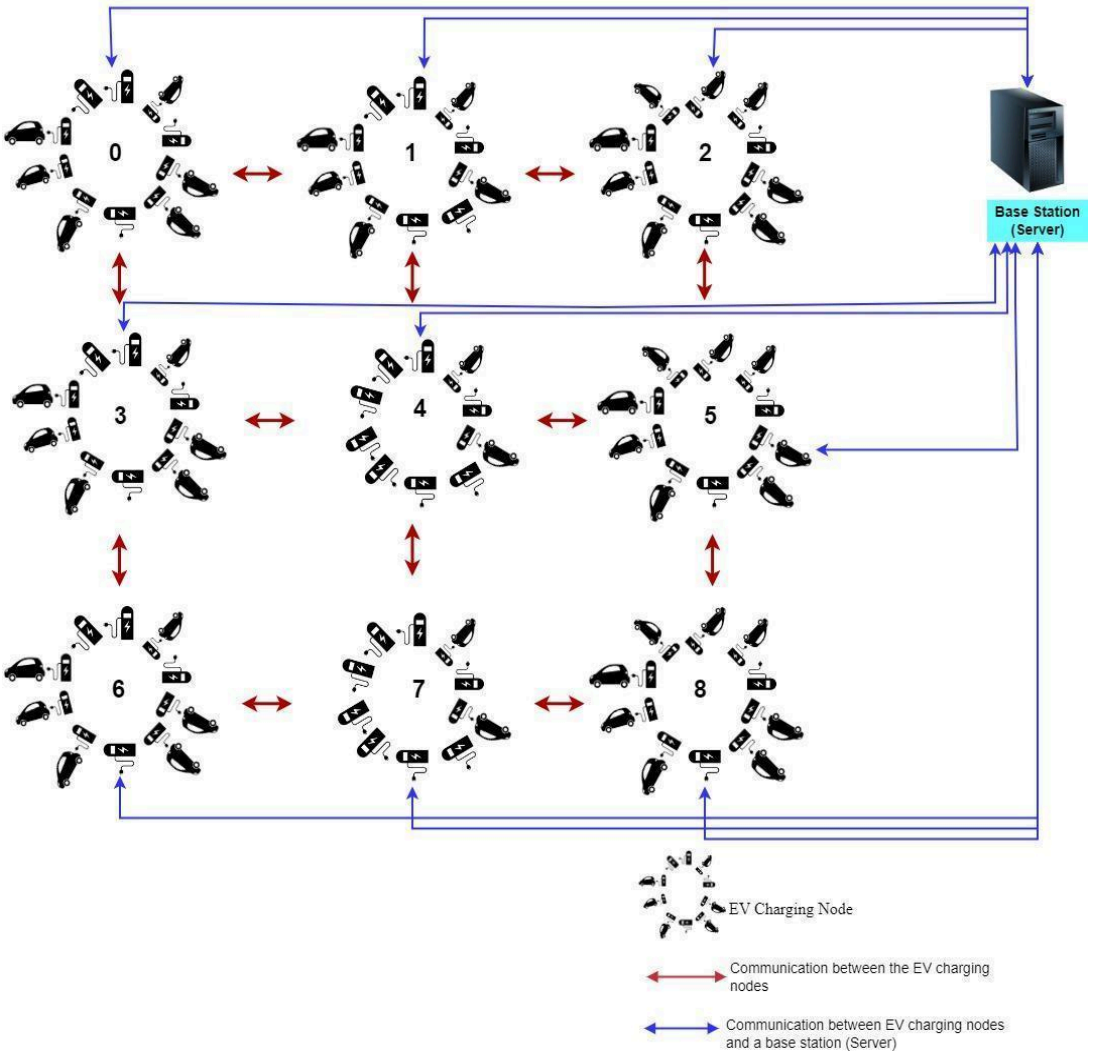


Figure 2. Overview of the wireless sensor network as the electric charging grid. In this illustration, the EV charging nodes are arranged in a 3 × 3 grid. Each EV charging node has a number of in-use or free charging ports. Each EV charging node communicates with the neighbours and the base station.

For simplicity, in this assessment, we assume the deployed wireless sensor network comprises **m × n** nodes in a Cartesian grid and a base station. In the example illustration above, m and n are both equal to 3. In your implementation, m and n are specified by the user, and your program should work with different values of m and n. Each node in the Cartesian grid simulates the behaviour of an EV charging node, and each node communicates with its immediate neighbours. These nodes can exchange data through both unicast and broadcast communication modes.

Each node in the WSN can independently exchange data with the base station. The base station for this WSN is an additional computer node responsible for gathering data from all charging nodes. All communication among nodes and between nodes and the base station is performed using the Message Passing Interface (MPI).

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Task 1 [EV charging nodes]

In this task, you are required to implement the EV charging nodes. In your implementation, you should use a single MPI process to simulate each EV charging node. As an example, there will be 9 MPI processes for a 3 × 3 Cartesian grid layout. Each node will have a (x, y) coordinate, e.g., Node 0 - (0,0), Node 1 - (0,1), Node 2 - (0,2), …, Node 8 (2, 2). In your implementation, you are required to use MPI virtual topologies.

Each EV charging node has ***k*** number of (in use/free) charging ports simulated with OpenMP or POSIX threads. Each charging port will periodically update its in-use/free status as a proxy to simulate the arrival and departure of EVs. The status information must be stored in a shared array accessible to all threads within the same node. You are allowed to set a fixed size for the array, provided that it can store all the necessary information.

The following table shows the sample charging ports availability of an EV charging node, where k=6 and simulated at a cycle of 10 seconds:

Table 1. Sample of charging ports availability for an EV charging node with 6 ports

Year	Month	Day	Hour	Minute	Second	Availability
2026	09	22	10	07	10	1
2026	09	22	10	07	20	6
2026	09	22	10	07	30	5
2026	09	22	10	07	40	2
2026	09	22	10	07	50	1

Note: You are free to set and test different update cycles, as long as it is realistic.

If all the ports of an EV charging node are full (or availability falls below a pre-set threshold), the node will request data from all neighbouring nodes. For a 2D Cartesian grid, the neighbouring nodes are the immediate top, bottom, right, and left nodes (if they exist). If the received data from the neighbours shows vacancies, the node shall ask incoming EV(s) to one of the available neighbouring nodes. However, if the received data also indicates that all ports are utilised (with availability below a pre-set threshold), the node will need to send a report alerting the base station to the congestion incident.

The report sent to the base station should contain as much information as possible. The report must include the timestamp (date & time) indicating when a congestion incident is discovered. The report should also include basic identification information for the reporting nodes, as well as port availability for the reporting node and its neighbouring nodes. The number of messages exchanged with the neighbour nodes and their message sizes should also be reported. After sending the report, the node is expected to receive information from the base station about the nearest available EV charging nodes, which the node will then forward to the EVs.

For this assessment, we assume the simulation will continue until all the EV charging nodes receive termination messages from the base station. Your implementation must be able to perform a "graceful shutdown", including successfully stopping all MPI processes, handling any unfinished I/O operations, and freeing all allocated memory without errors.

Reminder: To test your implementation, you should have low-availability test cases to verify that the data request is working and that communication can be sent to/from neighbouring nodes.

Reminder: You should demonstrate efficiency and reduce overhead when reporting to the base station. Are you using too many MPI message calls? Is your implementation wasting time while waiting for a blocking message call?

Note: If your implementation/simulation finished with system/memory errors (e.g., segmentation faults, bus errors), marks will be deducted.

Task 2 [Base station]

In this task, you are required to implement and simulate the base station node with *a single MPI* process. The base station node must periodically listen for incoming reports from the EV charging nodes. Once a report is received, the base station must check for the *nearest available neighbouring nodes* based on the reporting information. If there are more than 1 nearest neighbouring node, the base station is required to report *all the nearest nodes*.

Base station information update mechanism: Assume the reporting node is Node 0, and the neighbouring nodes are Node 1 and Node 3. The base station will check whether there are any reports received from Nodes 1 and 3 previously (which also cover information about Nodes 2, 4 and 6):

- If no report is received from at least one of the nodes within a predefined period, the base station will compute and suggest the available nodes (i.e., choosing the right candidates from Node 2, 4, and 6) to the reporting node.
- If reports are received from all nodes within a predefined period, the base station will send a message to the reporting node indicating that no nodes are available nearby.

Note: In your implementation, you should test with different predefined period values.

In your implementation, the base station should always write (or log) key performance metrics (including at least the simulation time, number of alerts detected, total number of messages/events, senders' report information, computed available nodes, and other crucial information) to an output file for every simulation run.

At the end of the simulation, the base station is also required to send a termination message to all the EV charging nodes, and is capable of performing a "graceful shutdown", including successfully stopping the MPI processes, handling all the unfinished I/O procedures, and freeing all the allocated memory without any errors.

Hint: The base station should avoid excessive overhead (e.g., thread creation/destruction) by using only one thread to send or receive MPI messages from the EV charging nodes.

Task 3 [Analysis]

After implementing Tasks 1 and 2, you are required to perform a communication analysis of your implemented simulation software.

For *each type* of message, you are required to measure the communication time (delay) and answer the following questions with simulation results.

a) How would the communication time (delay) scale when the EV grid size increases?

b) If the EV charging nodes are arranged in a line topology, how would the communication time (delay) scale as the number of nodes increases?

Task 4 [Presentation slides and documentation]

In this task, you are required to prepare presentation slides (or documentation) to report and present your design, answers, and results from previous Tasks. You will have a maximum of 7 minutes to present your slides and approximately 1 to 2 minutes for the Q&A session.

Your presentation slides (or documentation) should have the following sections and content (at minimum):

a) (Task 1) EV Charging Nodes

This section should present your approach to implementing Task 1. A clear diagram showing how you model the charging nodes and the charging ports (within each node) is required to be included in your presentation. How your software simulates the EV charging and what types of message passing are required should also be included in your presentation. You should also present the crucial part(s) of your code/pseudo-code to illustrate how you implement the task.

b) (Task 2) Base station

This section should present your approach to implementing Task 2. A clear diagram showing how you model the base station must be included in your presentation. How your software simulates the base station and what types of message passing are required should also be included in your presentation. You should also present the crucial part(s) of your code/pseudo-code to illustrate how you implement the task.

c) (Task 3) Analysis

This section should present the communication analysis of your implementation. You are also required to produce and present the following graphs (at a minimum) *for each type of message* used in your architecture design:

- Graphs showing how the communication delay would change when the EV grid size increases
- Graphs showing how the communication delay would change when the size of the line topology increases

d) Q&A
The section invites questions from the audience (TA/makers). The teaching team will ask questions related to your presentation and the submitted files.

Note: You are allowed to merge the (architecture) diagram for Task 1 and Task 2 if merging improves the presentation clarity.

Note: Practice makes perfect. If you are still deciding what to present and searching for information/content from the slides, chances are you will not be able to present within the time limit.

Note: You are allowed to put extra slides as an appendix (after the Q&A section) for your own convenience. Make sure they are clearly labelled so that markers will not be confused or mix them up with the other required sections.

Note: Please note that any materials (e.g., extra slides or documents) that are NOT properly presented (whether in the appendix or not) will generally not be marked. Make sure you allocate time wisely, plan your speech, and reserve enough time for every slide/document section.

Submission Checklist

- Task 1 and 2 – All C/C++ source codes, sample Base Station logs, and CAAS cluster scripts (if available)
- Task 3 – All time measurement logs and associated graphs (if available)
- Task 4 – Presentation slides (or documentation)
    - The documentation can be in the format of slides, docx, PDF or an equivalent format.
    - Note that we generally do not accept URLs or hyperlinks as submissions, as the content (slides/files/documents) referenced by the link is not guaranteed to remain unchanged after the deadline. To maintain fairness, markers may not mark content solely referenced via URLs or hyperlinks.
    - Please aim to keep the slides or documentation to within a 7-minute presentation.
    - Focus on emphasising the design, results and observations that would demonstrate a good understanding of the lab tasks.

- A separate AI declaration (in PDF format), **if not** already declared in the submitted documentation as aforementioned.

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INSTRUCTIONS FOR ALTERNATIVE ASSESSMENT

This section only applies if you or your team member(s) are **granted a special consideration extension** by the special consideration team (i.e., with a valid approval notice).

1. As a reminder, the regular 2-day extensions do not apply to applied/lab sessions (as they are in-class assessments with group/team work).

2. Instead of reviewing students' answers (in-class) in a presentation, peer review, and/or question-and-answer format during the regular in-class session, **students' answers (Task 4) will be assessed based on a recorded presentation video.** The video must be less than 9 minutes, and markers will assess the assessment offline.

3. Markers will only mark the first 9 minutes of the video at regular playback speed. If students decide to compress the video by speeding up playback, presentation quality may be affected, and marks may be deducted according to the marking rubric.

4. Instead of having in-person Q&A sessions, students will need to answer a set of fixed questions in their recording. The questions will be released in your special consideration letter. If you have special considerations approved but the letter does not include the questions, please contact our administrative staff: Jordan Nathanael (for the Clayton campus) and Dr Jessie Leong (for the Malaysia campus).

5. Apart from the recorded video file (mp4), students are still required to submit the presentation slides (PDF/pptx), associated source codes (if any), and any AI declarations/reports (PDF) listed in the regular submission checklist via Moodle before **the approved extended due date**.

6. **A late penalty** (5% per day) will still apply if you do not submit the required materials on time (against the new extended date)! As a reminder, if you submit 7 days after the extended due date, you will still receive a mark of 0.

7. AI tools are generally not allowed during the recording sessions.

8. Before you start your presentation, you are required to show your face and your student ID in the video.